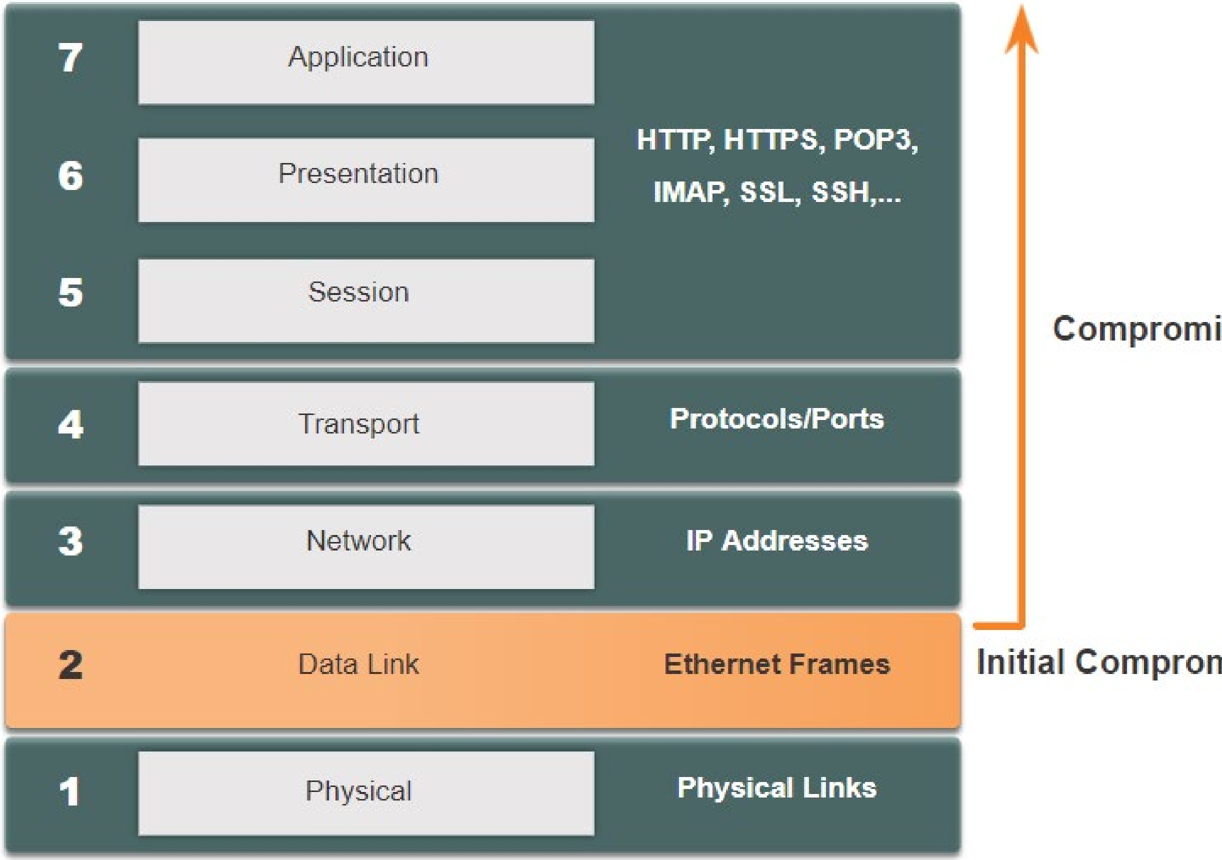
10.3 Layer 2 Security Threats

Layer 2 Vulnerabilities

If Layer 2 is compromised, then all the layers above it are also affected.

A threat actor with access to the internal network captured Layer 2 frames,

The threat actor could cause a lot of damage on the Layer 2 LAN networking infrastructure.



Switch Attack Categories

Security is only as strong as the weakest link in the system, and Layer 2 is considered to be that weak link. This is because LANs were traditionally under the administrative control of a single organization. We inherently trusted all persons and devices connected to our LAN. Today, with BYOD and more sophisticated attacks, our LANs have become more vulnerable to penetration.

Category	Examples
MAC Table Attacks	Includes MAC address flooding attacks.
VLAN Attacks	Includes VLAN hopping and VLAN double-tagging attacks. It also includes attacks between devices on a common VLAN.
DHCP Attacks	Includes DHCP starvation and DHCP spoofing attacks.
ARP Attacks	Includes ARP spoofing and ARP poisoning attacks.
Address Spoofing Attacks	Includes MAC address and IP address spoofing attacks.
STP Attacks	Includes Spanning Tree Protocol manipulation attacks.

Switch Attack Mitigation Techniques

Solution	Description
Port Security	Prevents many types of attacks including MAC address flooding attacks and DHCP starvation attacks.
DHCP Snooping	Prevents DHCP starvation and DHCP spoofing attacks.
Dynamic ARP Inspection (DAI)	Prevents ARP spoofing and ARP poisoning attacks.
IP Source Guard (IPSG)	Prevents MAC and IP address spoofing attacks.

These Layer 2 solutions will not be effective if the management protocols are not secured. The following strategies are recommended:

- Always use secure variants of management protocols such as SSH, Secure Copy Protocol (SCP), Secure FTP (SFTP), and Secure Socket Layer/Transport Layer Security (SSL/TLS).
- Consider using out-of-band management network to manage devices.
- Use a dedicated management VLAN where nothing but management traffic resides.
- Use ACLs to filter unwanted access.

10.4 MAC Address Table Attack

Switch Operation Review

Recall that to make forwarding decisions, a Layer 2 LAN switch builds a table based on the source MAC addresses in received frames. This is called a MAC address table. MAC address tables are stored in memory and are used to more efficiently switch frames.

```
S1# show mac address-table dynamic
          Mac Address Table
-------------------------------------------
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
   1    0001.9717.22e0    DYNAMIC     Fa0/4
   1    000a.f38e.74b3    DYNAMIC     Fa0/1
   1    0090.0c23.ceca    DYNAMIC     Fa0/3
   1    00d0.ba07.8499    DYNAMIC     Fa0/2
S1#
```

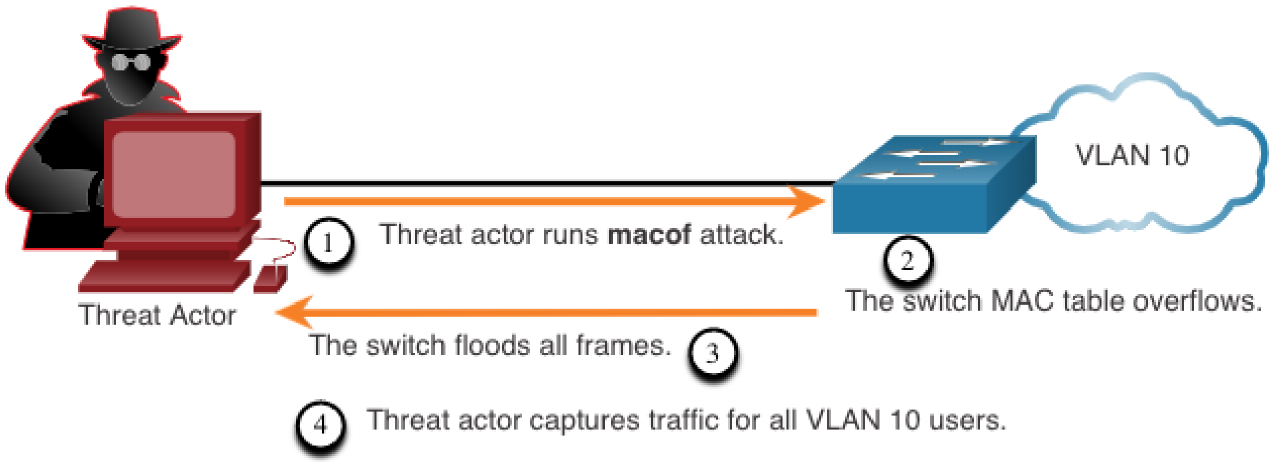
MAC Address Table Flooding

All MAC tables have a fixed size

The Switch can run out of resources in which to store MAC addresses.

MAC address flooding attacks - bombarding the switch with fake source MAC addresses until the switch MAC address table is full.

Note: Traffic is flooded only within the local LAN or VLAN. The threat actor can only capture traffic within the local LAN or VLAN to which the threat actor is connected.



MAC Address Table Attack Mitigation

- **Attack tool macof**
- Attacker can create a MAC table overflow attack very quickly.
- For instance, a Catalyst 6500 switch can store 132,000 MAC addresses in its MAC address table.
- A tool such as **macof** can flood a switch with up to 8,000 bogus frames per second; creating a MAC address table overflow attack in a matter of a few seconds.
- Also affect other connected Layer 2 switches.
- When the MAC address table of a switch is full, it starts flooding out all ports including those connected to other Layer 2 switches.
- To mitigate MAC address table overflow attacks, network administrators must implement port security.
- Port security will only allow a specified number of source MAC addresses to be learned on the port. Port security is further discussed in another module.